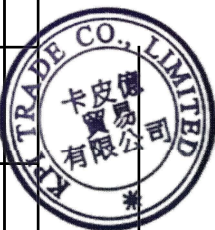


KPI TRADE CO., LIMITED
MATERIAL CERTIFICATE(EN 102043.1)

Certificate No.:XZ2501
Customer: Valvtronic S.A.
Material grade:ASTM A-105/A-105M-23

Date: JAN 5 2025.
PO.:28082
Heat Treatment:Normalized (920°C), holding time 180 min, air cooling.

No.	Description			Q'TY	Heat No.	Chemical Composition (%)													
						C	Mn	Si	P	S	AL	Cu	Ni	Cr	Mo	V			
430	3"	#600	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
450	3"	#300	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
480	3"	#300	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
510	3"	#300	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
530	3"	#600	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
540	3"	#150	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
550	4"	#150	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
560	4"	#600	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
590	4"	#300	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
630	4"	#300	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
640	4"	#600	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
680	4"	#300	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
690	4"	#600	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
700	4"	#150	RF	A05714	0.19	0.88	0.19	0.014	0.006	0.039	0.010	0.002	0.001	0.002	0.001				
710	6"	#150	RF	A02196	0.19	0.86	0.2	0.016	0.007	0.032	0.030	0.004	0.001	0.002	0.002				
740	6"	#300	RF	A02196	0.19	0.86	0.2	0.016	0.007	0.032	0.030	0.004	0.001	0.002	0.002				
810	6"	#300	RF	A02196	0.19	0.86	0.2	0.016	0.007	0.032	0.030	0.004	0.001	0.002	0.002				
Heat Treatment Batch NO.				Tension Test			Hardness Test (HB)			Charpy Impact Test			NDE						
				Y.S Mpa		T.S Mpa	E.L %	R.A %	Temp(°C)			Impact value		UT		MT		PT	
				Min		Max.	197	148											
A05714								/		/		OK		N/A		OK			
A02196								/		/		OK		N/A		OK			



Q.A Dept.